

PlatformIO Installation & Setup Guide (Shakti – Ganga/Yamuna)

This guide explains how to install PlatformIO, configure it for Shakti SoCs (Ganga/Yamuna), and run a sample Hello World program.

1. Install PlatformIO in VSCode

Install Python virtual environment support (required by PlatformIO):
Open terminal and run:

```
sudo apt install python3-venv
```

- Install PlatformIO IDE in VSCode
- Open VSCode
- Go to Extensions (left sidebar)
- Search: PlatformIO IDE
- Click Install

VSCode will reload automatically after installation

2. Clone the GC2025 Repository

Open the terminal and run the below command

```
git clone https://gitlab.com/shaktiproject/gc2025.git
```

If already cloned go to the directory and then run:

```
git pull
```

Inside this repository, you will find the script named [.platformio.sh](#)
cd <location where repo is cloned>/platform.sh

2.1 Run the Script:

Open the terminal and run:

```
chmod +x .platformio.sh
```

```
./platformio.sh
```

The script automatically installs:

- Shakti Platform → ~/.platformio/platforms/shakti-gc2025
- Shakti SDK (framework) →
~/.platformio/packages/framework-shakti-gc2025-sdk
- Shakti OpenOCD → ~/.platformio/packages/shakti-openocd

3. Create & Run Hello World on Ganga/Yamuna

- Create a New PlatformIO Project
- Open VSCode
- Click PIO Home (PlatformIO icon on left panel)
- Select New Project

Configure:

- Name your_project_name
- Board ganga or yamuna
- Framework shakti-gc2025-sdk

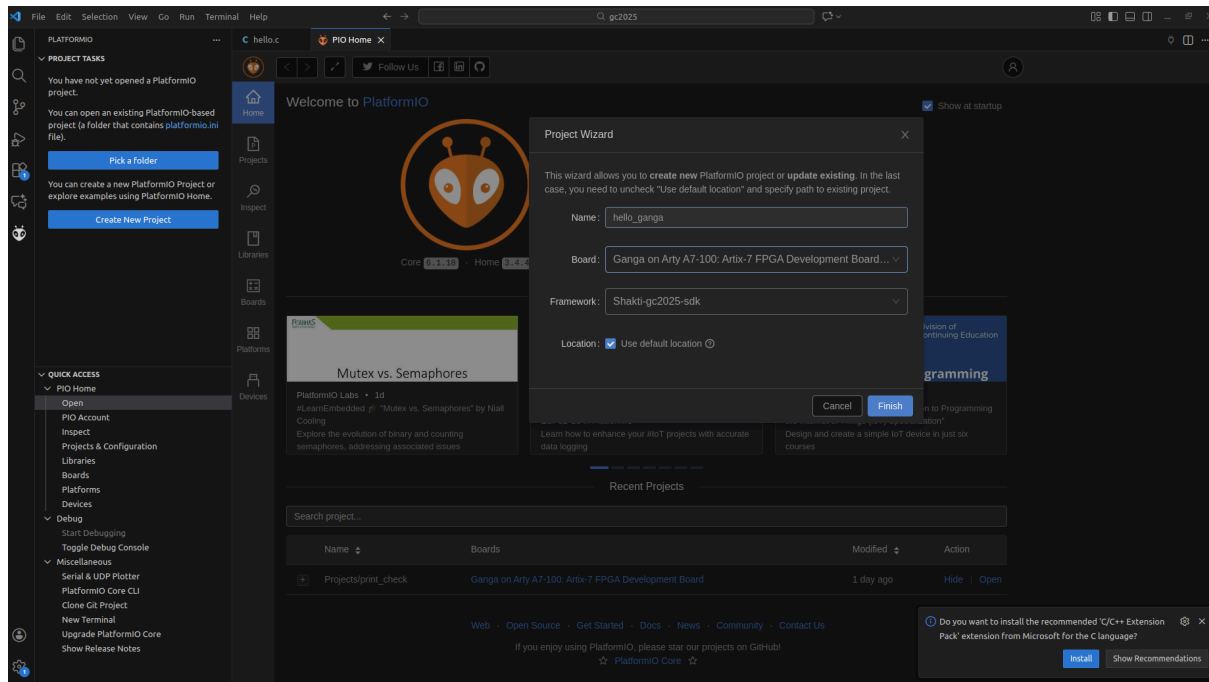


Fig 1: To create a project

Create the project

- Add Hello World Code
- Create a file:
src/main.c

Paste:

```
#include <stdio.h>
```

```
int main() {  
    printf("Hello World from Shakti!\n");  
    while (1);  
    return 0;  
}
```

4. View Output Using Serial (miniterm)

- Open a new terminal inside VSCode:

```
pyserial-miniterm /dev/ttyUSB0 19200
```

Replace USB0 with your device's usb port.
Baud: 19200 (default for Shakti UART)

5. Build & Upload

Run the following tasks:

- Build
- Upload

After upload completes, Hello World output should appear in miniterm

6. Debugging in PlatformIO

PlatformIO uses OpenOCD + GDB to debug Shakti boards. When you start debugging, PlatformIO automatically loads the firmware.

6.1 Start the Debugger

- Click Run & Debug (left sidebar)
- Select PIO Debug (gdb)
- The debugger:
 1. Connects to FTDI
 2. Loads the program
 3. Halts at main()

You will see:

Temporary breakpoint 1, main() at src/main.c:4

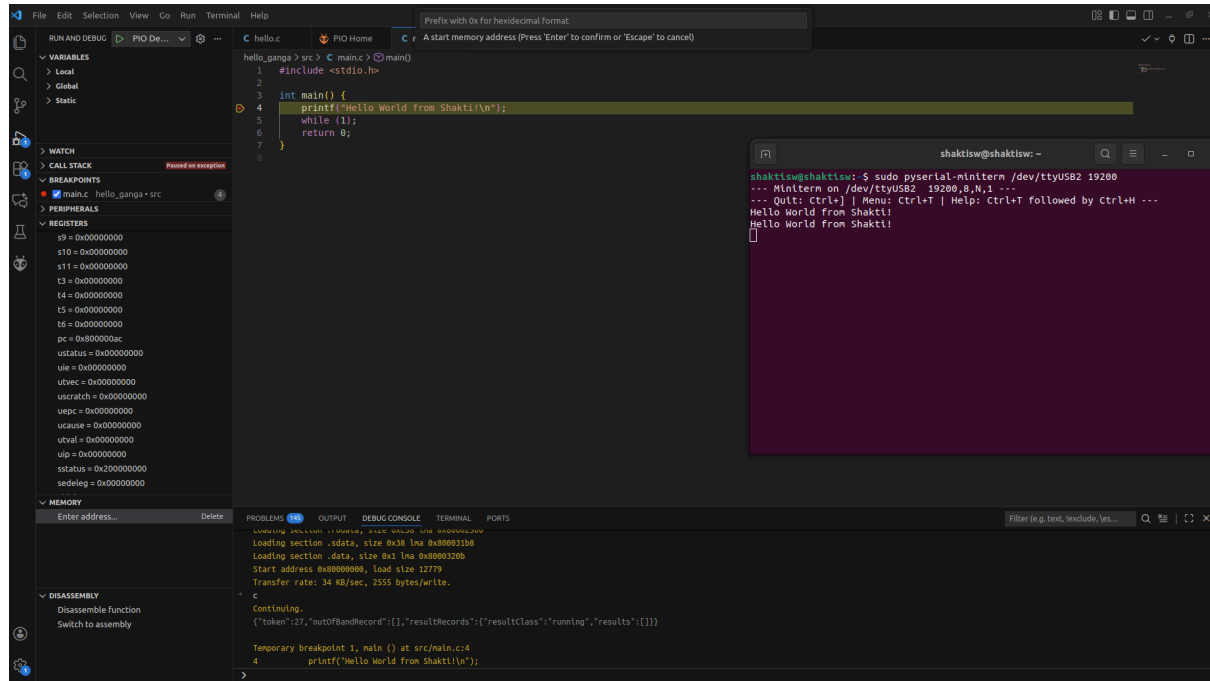


Fig 2: Temporary breakpoint

PlatformIO: Initialization completed

6.2 Breakpoints

- Click to the left of a line number to place a breakpoint
- Example: Set a breakpoint on the printf() line
- Execution will stop exactly at that line when you run the debugger.

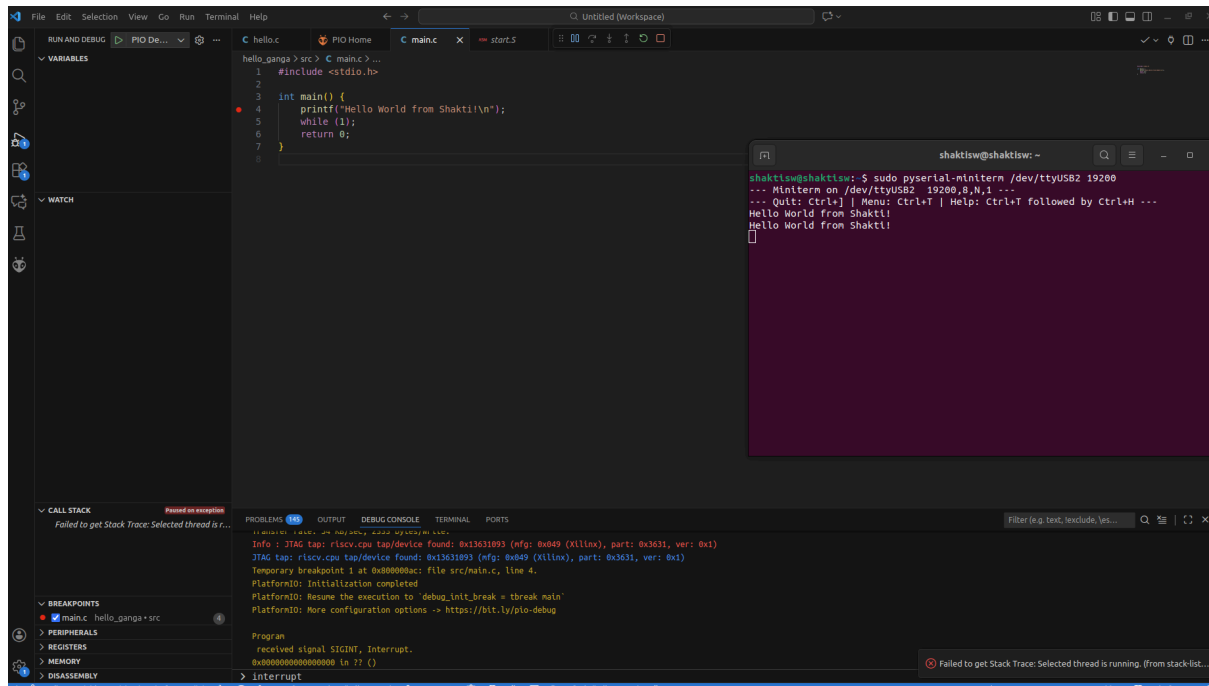


Fig 3: To set a breakpoint

6.3 Debug Controls

Button Function

1.  Continue Run program from current point
2.  Step Over Execute next line (don't enter functions)
3.  Step Into Enter a function call
4.  Step Out Return from current function
5.  Restart Reload and restart debugging
6.  Stop End debugging session

6.4 Debug Windows (What You See in Fig 2)

- Variables → Shows current local/global variables
- Registers → Shows RISC-V register values
- Peripherals → Hardware register view
- Memory → Enter an address (e.g., 0x80000000) to inspect memory and number of address locations that are to be checked.
- Call Stack → Shows current function call hierarchy